

Perimeter of

Irregular Shapes

Becoming a Boundary Explorer

The Missing Border Mystery

You are helping decorate the classroom for a festival.

You have a beautiful poster — but it's missing its colorful border!

You have a roll of ribbon, but you don't want to waste it.

How do you find out exactly how much ribbon you need to go all the way around?

Quick Question

If you walked along the edge of your school desk, what would you be measuring?

The top surface OR the boundary?

What Is Perimeter?

Definition: Perimeter is the total distance around the outside of a 2D shape.

The Ant Walk Method

Imagine a tiny ant starting at one corner of a shape.
The ant walks exactly once all the way along the edge
until it returns to the same corner.

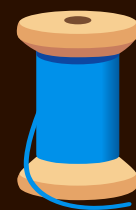
The total distance the ant traveled = the Perimeter!

Perimeter Around Us in India 🇮🇳



Rangoli Border

During Diwali, a border of white powder or petals around a Rangoli — the length of that border is the perimeter!



Gota Lace Border

Putting Gota lace on a Dupatta or Saree? You need the perimeter of the cloth to buy the right length of lace!



Colony Garden Fence

To keep animals out of the park, a fence is put up. The total length of that fence is the perimeter of the park.

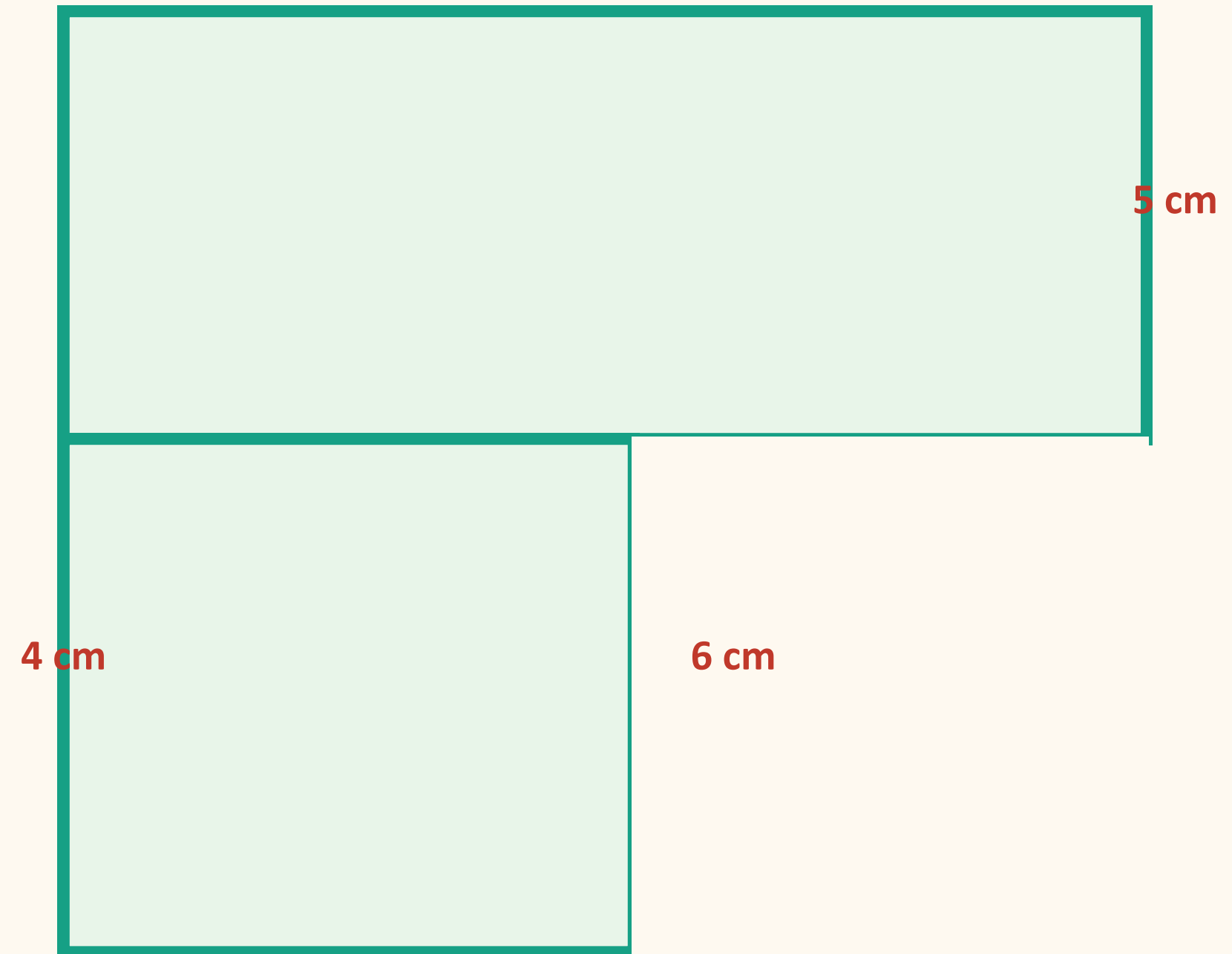
Irregular Shapes: Not All Sides Are Equal!

Regular shapes like squares and rectangles follow strict rules — all sides equal.

Irregular shapes have sides of different lengths and angles of different sizes.

Key Fact

Even if a shape looks strange or has many corners, we can still find its perimeter. We just need to be very careful to count every single side!



An Irregular (L-shaped) Figure

How to Find the Perimeter

Perimeter = **Sum of ALL the side lengths**

Example: A 4-Sided Shape

Sides: 4 cm, 5 cm, 3 cm, and 6 cm

$$4 + 5 + 3 + 6 = 18 \text{ cm}$$



Explorer's Tip

Always cross out or mark the sides you have already added so you don't count the same side twice or miss one!

Practical Activity: The L-Shaped Garden

Walk the Boundary — Sides: 5, 7, 4, 6, 3, 5 cm

Step 1

$$5 + 7 = 12$$

Start at the top

Step 2

$$12 + 4 = 16$$

Add the next side

Step 3

$$16 + 6 = 22$$

Keep going...

Step 4

$$22 + 3 = 25$$

And the next...

Step 5

$$25 + 5 = 30 \text{ cm } \checkmark$$

Finally the last side

Total Perimeter = 30 cm

Why Add Every Side?

 In irregular shapes, students often forget the small 'turns' or 'inner corners.'

The Missing Gap Rule

If you forget to add even one side, your boundary will have a **gap!**

Imagine a fence with a missing piece — the animals would get right in. 

To find the true perimeter, **every side must form part of the sum.**

Interactive Practice: Rangoli Borders

Can you find the perimeter for each design?

Design A — T-Shape

Sides: 10, 2, 4, 6, 4, 2 cm

Your Calculation:

Perimeter = _____ cm

Answer: $10+2+4+6+4+2 = 28$ cm

Design B — Jagged Shape

Sides: 5 cm $\times$ 8 sides

Your Calculation:

Perimeter = _____ cm

Answer: $5 \times 8 = 40$ cm

The Ant Walk Mental Check

You have mastered the perimeter of irregular shapes!

✓ Perimeter is the total distance around a shape.

✓ For Irregular Shapes, not all sides are equal — but we still add them all.

✓ Always Walk the Boundary from start to finish.

✓ The total distance is the Sum of every single side.

Next time you see a Rangoli or a garden, try to 'walk the boundary' in your mind!



Thank You!

Keep Exploring Boundaries!

